STATS 413 — Midterm 2 Reference Sheet

THE LINEAR MODEL (L1–4)

Weaker: $\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $\text{Var}(\boldsymbol{\varepsilon}) = \sigma_\varepsilon^2 I$. **Stronger:** also $\boldsymbol{\varepsilon} \sim \text{MVN}(\mathbf{0}, \sigma_\varepsilon^2 I)$. X: $n \times (p+1)$, first col $= \mathbf{1}$.

Random Vector Properties

$E(A\boldsymbol{Z}+\boldsymbol{b}) = AE(\boldsymbol{Z})+\boldsymbol{b}$, $\text{Var}(A\boldsymbol{Z}+\boldsymbol{b}) = A\text{Var}(\boldsymbol{Z})A^\top$, $\text{Var}(\boldsymbol{Z}) = E[\boldsymbol{Z}\boldsymbol{Z}^\top] - E[\boldsymbol{Z}]E[\boldsymbol{Z}]^\top$

What Each Model Gives Us

Weaker: unbiased $\hat{\boldsymbol{\beta}}$, $\text{Var}(\hat{\boldsymbol{\beta}})$, Gauss-Markov. **Stronger adds:** tests, CIs, partial F, PI.

OLS ESTIMATION (L2–4)

$$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \boldsymbol{y} = \arg\min_{\tilde{\boldsymbol{\beta}}} \|\boldsymbol{y} - X\tilde{\boldsymbol{\beta}}\|^2$$

Linear in $\boldsymbol{y}$: $\hat{\boldsymbol{\beta}} = A\boldsymbol{y}$, $A = (X^\top X)^{-1} X^\top$.

Under Weaker Model

$E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ (unbiased), $\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma_\varepsilon^2 (X^\top X)^{-1}$

$\text{Var}(\hat{\beta}_j) = \sigma_\varepsilon^2 [(X^\top X)^{-1}]_{(j+1),(j+1)}$

Under Stronger Model

$\hat{\boldsymbol{\beta}} \sim \text{MVN}\big(\boldsymbol{\beta},\ \sigma_\varepsilon^2 (X^\top X)^{-1}\big)$

$\hat{\beta}_j \sim N\big(\beta_j,\ \sigma_\varepsilon^2 [(X^\top X)^{-1}]_{(j+1),(j+1)}\big)$

MLE: Under normality, MLE of $\boldsymbol{\beta}$ = OLS $\hat{\boldsymbol{\beta}}$.

Gauss-Markov (BLUE): Under weaker, OLS has smallest MSE among linear unbiased est. Key: any linear $\tilde{\boldsymbol{\beta}} = A\boldsymbol{y}$ with $A = (X^\top X)^{-1} X^\top + D$, $DX = \mathbf{0} \Rightarrow \text{Var}(\tilde{\boldsymbol{\beta}}) = \text{Var}(\hat{\boldsymbol{\beta}}) + \sigma_\varepsilon^2 DD^\top$.

HAT MATRIX & PROJECTIONS (L4, 10)

$H = X(X^\top X)^{-1} X^\top$, $\hat{\boldsymbol{y}} = H\boldsymbol{y}$, $\boldsymbol{e} = (I - H)\boldsymbol{y}$

Key Matrix Identities

- $H^\top = H$, $H^2 = H$ (symmetric, idempotent)
- $(I-H)^\top = I-H$, $(I-H)^2 = I-H$
- $H(I-H) = \mathbf{0}$, $X^\top \boldsymbol{e} = \mathbf{0}$
- H projects onto $\text{col}(X)$; $(I-H)$ onto $\text{col}(X)^\perp$

Diagonal Elements

$1/n \le h_{ii} \le 1$, $\sum h_{ii} = p+1$, $\hat{y}_i = h_{ii} y_i + \sum_{j \ne i} h_{ij} y_j$

Under Weaker Model

$E(\boldsymbol{e}) = \mathbf{0}$, $E(\hat{\boldsymbol{y}}) = X\boldsymbol{\beta}$. $\text{Var}(\boldsymbol{e}) = \sigma_\varepsilon^2 (I-H)$, $\text{Var}(\hat{\boldsymbol{y}}) = \sigma_\varepsilon^2 H$

$\text{Var}(e_i) = \sigma_\varepsilon^2 (1-h_{ii})$, $\text{Var}(\hat{y}_i) = \sigma_\varepsilon^2 h_{ii}$, $\text{Cov}(e_i, \hat{y}_j) = 0$, $\text{Cov}(e_i, e_j) = -\sigma_\varepsilon^2 h_{ij}$

Stronger: $\boldsymbol{e} \perp\!\!\!\perp \hat{\boldsymbol{y}}$. $e_i \sim N(0, \sigma_\varepsilon^2 (1-h_{ii}))$. If $h_{ii} = 1$: $\hat{y}_i = y_i$, $e_i = 0$.

Algebraic (w/ intercept): $\bar{e} = 0$, $\text{cov}(e, \hat{y}) = 0$, $\text{cov}(e, x_j) = 0$

R^2 AND SUMS OF SQUARES (L4)

$\text{TSS} = \sum (y_i - \bar{y})^2$, $\text{RSS} = \sum e_i^2$, $\text{SSreg} = \sum (\hat{y}_i - \bar{y})^2$. TSS = SSreg + RSS (w/ intercept).

$R^2 = 1 - \text{RSS}/\text{TSS} = \text{cor}(\hat{y}, y)^2$. Simple: $R^2 = \text{cor}(x, y)^2$. Never decreases adding predictors.

ESTIMATING σ_ε (L5)

$\hat{\sigma}_\varepsilon^2 = \text{RSS}/(n-p-1)$. df $= n-p-1$ ($p+1$ constraints from $\boldsymbol{e}^\top X = \mathbf{0}$).

$\text{se}(\hat{\beta}_j) = \hat{\sigma}_\varepsilon \sqrt{[(X^\top X)^{-1}]_{(j+1),(j+1)}}$

HYPOTHESIS TESTS (L5–6)

t-test for β_j

$H_0 : \beta_j = c$. Test statistic:

$$t = \frac{\hat{\beta}_j - c}{\text{se}(\hat{\beta}_j)} \sim t_{n-p-1} \quad \text{under } H_0$$

$\ne c$: $p = 2P(T \ge |t|)$. $> c$: $P(T \ge t)$. $< c$: $P(T \le t)$. Reject if $p \le \alpha$.

CI for β_j

$$\hat{\beta}_j \ \pm\ t_{1-\alpha/2,\ n-p-1} \cdot \text{se}(\hat{\beta}_j)$$

Duality: reject $H_0 : \beta_j = c$ at $\alpha \iff c \notin$ the $100(1-\alpha)\%$ CI.

Overall F-test

$H_0 : \beta_1 = \cdots = \beta_p = 0$, H_a: at least one $\ne 0$.

$$F = \frac{R^2/p}{(1-R^2)/(n-p-1)} = \frac{(\text{TSS}-\text{RSS})/p}{\text{RSS}/(n-p-1)} \sim F_{p,\ n-p-1}$$

p-value $= P(F_{p,n-p-1} \ge F_{\text{obs}})$. Always right-tail.

Partial F-test

Full vs. reduced (nested). H_0: subset of β's $= 0$.

$$F = \frac{(\text{RSS}_R - \text{RSS}_F)/(df_R - df_F)}{\text{RSS}_F/df_F} \sim F_{df_R - df_F,\ df_F}$$

Num. df $= p_F - p_R$. Overall F: reduced = intercept only.

Key: $t_{df}^2 = F_{1,df}$ — partial F for one predictor = t-test.

CATEGORICAL VARIABLES (L7)

G levels $\to G-1$ **dummy variables**. Omitted = **reference**.

$\hat{\beta}_0 = \bar{y}_{\text{ref}}$, $\hat{\beta}_j = \bar{y}_j - \bar{y}_{\text{ref}}$ (one categorical only).

Interpretation: $\hat{\beta}_j$ = est. diff. in $E(y)$ between cat. j and reference, *holding other vars fixed*. Why not G? $\sum$ dummies $= \mathbf{1}$ $\Rightarrow X^\top X$ singular.

One categorical + overall F = **ANOVA**: $H_0 : \mu_1 = \cdots = \mu_G$. Adding continuous predictors: slopes compare groups *holding continuous vars fixed*.

INTERACTIONS (L7–8)

Cat × Cont: $x_i \cdot \mathbf{1}\{z = c\}$ gives group-specific slopes. Full interactions = separate regressions per group.

Cont × Cont: include $x_1 x_2$ as predictor:

$$\frac{\partial \hat{y}}{\partial x_1} = \hat{\beta}_1 + \hat{\beta}_{12}\, x_2$$

Interpreting: slope in subgroup $= \hat{\beta} + \hat{\beta}_{\text{interaction}}$. Define $\boldsymbol{a}$ s.t. $\boldsymbol{a}^\top \hat{\boldsymbol{\beta}}$ = desired combination.

Simpson's Paradox: Aggregated trend reverses within subgroups when confounder is ignored.

LINEAR COMBINATIONS (L8)

For any fixed vector $\boldsymbol{a}$:

$$E(\boldsymbol{a}^\top \hat{\boldsymbol{\beta}}) = \boldsymbol{a}^\top \boldsymbol{\beta} \qquad \text{Var}(\boldsymbol{a}^\top \hat{\boldsymbol{\beta}}) = \sigma_\varepsilon^2\, \boldsymbol{a}^\top (X^\top X)^{-1} \boldsymbol{a}$$

$$\text{se}(\boldsymbol{a}^\top \hat{\boldsymbol{\beta}}) = \hat{\sigma}_\varepsilon \sqrt{\boldsymbol{a}^\top (X^\top X)^{-1} \boldsymbol{a}}$$

Cannot compute from `summary()` alone—need $\hat{\sigma}_\varepsilon$ and X.

CONDITIONAL EXPECTATIONS (L9)

$\hat{\mu}_{y|\tilde{\boldsymbol{x}}} = \tilde{\boldsymbol{x}}^\top \hat{\boldsymbol{\beta}}$ estimates $\mu_{y|\tilde{\boldsymbol{x}}} = E(y|\tilde{\boldsymbol{x}}) = \tilde{\boldsymbol{x}}^\top \boldsymbol{\beta}$

$$E(\hat{\mu}_{y|\tilde{\boldsymbol{x}}}) = \mu_{y|\tilde{\boldsymbol{x}}} \qquad \text{Var}(\hat{\mu}_{y|\tilde{\boldsymbol{x}}}) = \sigma_\varepsilon^2\, \tilde{\boldsymbol{x}}^\top (X^\top X)^{-1} \tilde{\boldsymbol{x}}$$

Stronger: $\hat{\mu}_{y|\tilde{\boldsymbol{x}}} \sim N(\mu_{y|\tilde{\boldsymbol{x}}}, \sigma_\varepsilon^2 \tilde{\boldsymbol{x}}^\top (X^\top X)^{-1} \tilde{\boldsymbol{x}})$

CI for $\mu_{y|\tilde{\boldsymbol{x}}}$:

$$\hat{\mu}_{y|\tilde{\boldsymbol{x}}} \pm t_{1-\alpha/2,\, n-p-1} \cdot \hat{\sigma}_\varepsilon \sqrt{\tilde{\boldsymbol{x}}^\top (X^\top X)^{-1} \tilde{\boldsymbol{x}}}$$

Test $H_0 : \mu_{y|\tilde{\boldsymbol{x}}} = \mu_0$: $t = (\hat{\mu}_{y|\tilde{\boldsymbol{x}}} - \mu_0)/[\hat{\sigma}_\varepsilon \sqrt{\tilde{\boldsymbol{x}}^\top (X^\top X)^{-1} \tilde{\boldsymbol{x}}}] \sim t_{n-p-1}$

Variance **smallest** at $\tilde{\boldsymbol{x}} = \bar{\boldsymbol{x}}$; increases away from center.

PREDICTION INTERVALS (L9–10)

Future observation: $y^* = \tilde{\boldsymbol{x}}^\top \boldsymbol{\beta} + \varepsilon^*$, ε^* indep. of $\varepsilon_1, \ldots, \varepsilon_n$.

Two sources: estimation $\text{Var}(\hat{\mu}) = \sigma_\varepsilon^2 \tilde{\boldsymbol{x}}^\top (X^\top X)^{-1} \tilde{\boldsymbol{x}}$ + noise $\text{Var}(\varepsilon^*) = \sigma_\varepsilon^2$.

$$\text{Var}(y^* - \hat{\mu}_{y|\tilde{\boldsymbol{x}}}) = \sigma_\varepsilon^2 \big(1 + \tilde{\boldsymbol{x}}^\top (X^\top X)^{-1} \tilde{\boldsymbol{x}}\big)$$

Prediction Interval:

$$\hat{\mu}_{y|\tilde{\boldsymbol{x}}} \pm t_{1-\alpha/2,\, n-p-1} \cdot \hat{\sigma}_\varepsilon \sqrt{1 + \tilde{\boldsymbol{x}}^\top (X^\top X)^{-1} \tilde{\boldsymbol{x}}}$$

PI vs CI

- PI **always wider** (extra "1+" inside $\sqrt{\ }$)
- CI width $\to 0$ as $n \to \infty$; PI width $\to 2z\sigma_\varepsilon$
- CI targets $E(y|\tilde{\boldsymbol{x}})$; PI targets individual y^*
- Even knowing $\boldsymbol{\beta}$ perfectly, can't predict y^* perfectly

Requires: linearity (center), homosked. + normality (width). Homosked. lets us pool residuals to estimate σ_ε at any $\tilde{\boldsymbol{x}}$.

Prediction Error Decomposition

$E[(y^* - \tilde{\boldsymbol{x}}^\top \hat{\boldsymbol{\beta}})^2] = \text{Var}(\varepsilon^*) + \text{Var}(\tilde{\boldsymbol{x}}^\top \hat{\boldsymbol{\beta}})$. Cross term = 0 by independence.

DIAGNOSTICS (L10–11)

Internally Studentized Residuals

$r_i = e_i/[\hat{\sigma}_\varepsilon \sqrt{1-h_{ii}}]$. Under stronger: $\text{Var}(e_i/\sqrt{1-h_{ii}}) = \sigma_\varepsilon^2$. Use r_i for plots (approx. identically distributed).

1. Linearity

Plot e_i vs. $\hat{y}_i$ or each x_j. Good: amorphous blob. Bad: curvature.

2. Homoskedasticity

Plot r_i or $\sqrt{|r_i|}$ vs. $\hat{y}_i$. Good: constant spread. Bad: fanning.

3. Normality (QQ-plot of r_i)

Sort $r_{[i]}$, plot vs. $\Phi^{-1}(i/(n+1))$.

- On diagonal $\to$ normal
- Convex (cup) $\to$ right-skewed
- Concave $\to$ left-skewed
- S-shape $\to$ heavy tails

Consequences of Violations

Nonlinearity: biased predictions. Heterosked.: wrong SEs, p-values, CI/PI widths. Non-normality: t/F distributions invalid, coverage $\neq 1-\alpha$.

LEVERAGE (L11)

$h_{ii} = \boldsymbol{x}_i^\top (X^\top X)^{-1} \boldsymbol{x}_i$. Measures distance from $\bar{\boldsymbol{x}}$. Depends **only on** X.

$1/n \le h_{ii} \le 1$. Min at $\boldsymbol{x}_i = \bar{\boldsymbol{x}}$. Unique cat. level $\Rightarrow h_{ii} = 1$.

$\text{Corr}(\hat{y}_i, y_i \mid x_i) = \sqrt{h_{ii}}$. High leverage = *potential* influence, not necessarily influential. **Rule:** $h_{ii} > 2(p+1)/n$.

OUTLIER DETECTION (L12)

High h_{ii} + outlying $y_i \Rightarrow$ small e_i (line pulled toward it). Fix: refit without i.

Ext. studentized: $t_i = r_i \sqrt{(n-p-2)/(n-p-1-r_i^2)} \sim t_{n-p-2}$

Single: reject if $|t_i| > t_{1-\alpha/2,\, n-p-2}$. **All n:** Bonferroni at α/n.

Don't remove unless data error. R: `rstudent(lm)`

COOK'S DISTANCE (L12)

$D_i = \frac{1}{p+1} r_i^2 \frac{h_{ii}}{1-h_{ii}}$. Combines residual + leverage. **Rule:** $D_i > 1$ influential. R: `cooks.distance(lm)`

MULTICOLLINEARITY (L12–13)

Predictors linearly related $\Rightarrow X^\top X$ nearly singular. **Consequences:** unstable $\hat{\boldsymbol{\beta}}$, inflated SEs, weird signs. **Hallmark:** sig. overall F, no sig. t-tests.

VIF

$\text{VIF}_j = 1/(1-R_j^2)$. R_j^2 from regressing x_j on other predictors. Rule: > 10 concerning. R: `vif(lm)`

Condition Number

$\kappa = \sqrt{\lambda_{\max}/\lambda_{\min}}$ of predictor corr. matrix. > 15: moderate. > 30: severe.

Partial Regression

Regress y on $X_R \to y_R$; regress x_j on $X_R \to x_{j,R}$. Slope of y_R on $x_{j,R} = \hat{\beta}_j$. Pairwise: flag $|r| > 0.9$.

MULTIPLE HYP. TESTS (L13)

K tests at α: $P(\ge 1 \text{ Type I error}) \le K\alpha$. Under indep.: $1-(1-\alpha)^K$.

FWER $= P(\text{reject} \ge 1 \text{ true } H_0)$.

Bonferroni

Each test at $\alpha/K \Rightarrow$ FWER $\le \alpha$. Controls error but **reduces power**.

Holm–Bonferroni (uniformly better)

Order p-values $p_1 \le \cdots \le p_K$. Step i: reject if $p_i \le \alpha/(K+1-i)$; else stop. Rejects $\ge$ as many as Bonferroni.

Post-hoc Pairwise

After overall F rejects $\mu_1 = \cdots = \mu_G$: test $\binom{G}{2}$ pairs with Holm. Tells you **which** groups differ.

QUICK REFERENCE

Which Test?

Single β_j=0: t-test. Subset = 0: partial F. All slopes = 0: overall F. Group means equal: ANOVA. Outlier: ext. stud. t_i. Multiple: Holm.

Key R Commands

- `lm(y ~ x1 + x2, data)` — fit model
- `summary(lm)` — coefs, SEs, t-tests, R^2, F
- `confint(lm, level=0.95)` — CIs for β_j
- `anova(reduced, full)` — partial F-test
- `predict(lm, newdata, interval="confidence")`
- `predict(lm, newdata, interval="prediction")`
- `hatvalues(lm)` — h_{ii}; `rstandard(lm)` — r_i
- `rstudent(lm)` — t_i; `cooks.distance(lm)` — D_i
- `plot(lm, which=1:6)` — diagnostic plots
- `vif(lm)` — VIF (requires `car` package)
- `p.adjust(pvals, method="holm")` — Holm correction
- `model.matrix(lm)` — design matrix X